REFLOWECONOMY · FEASIBILITY ASSESSMENT

ReflowEconomy feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First feasibility assessment (kit 1.8 pipeline) of the reference micro-factory (option A)

ReflowEconomy feasibility assessment

The reference micro-factory is technically feasible on paper with open machines and workshop fabrication, and its science checks out (RFE-STM-001), but two things decide whether a real site works: the electrical supply, which as designed is too small once motor currents are counted (RFE-DEC-001 D1), and the price its products fetch, which has to be about 2.49 USD/kg for the site to break even. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Layout, zones, hoods, guarding and passport are complete at TRL 3 and constructable; every part is modelled at D3 with its fixings and passes the fidelity check with no interference	High	Hot zone clearance exactly at the 1.0 m limit; cooling press plates 0.9 m for a 1 m mould (to confirm when bought)	RFE-STM-001; RFE-CAL-001 sections 7 and 8; `cad/fidelity.json`
Materials	Standard stock (angle, SHS, RHS, flat bar, sheet, mesh, plywood), galvanised duct, mineral-wool panels, bought machines	High	Recycled feedstock quality varies with the input; PVC must be removed by hand at sorting	RFE-BLD-001; `bom/bom.csv`
Manufacturing and assembly	Eleven made components with making sketches; machines built to published open designs or bought; container seam only to a structural engineer's design	Medium	The derated 5 kW single-phase sheet press is not a published design; container wall removal needs an engineer and a certified welder	RFE-BLD-001; RFE-DDR-003
Power and controls	On paper the morning loads draw about 55 A from the 40 A single-phase supply; a heater interlock exists but cannot fix motor currents	Low	Fixes change a decided input: three-phase (recommended), a longer day in three blocks, or less input (RFE-DEC-001 D1); interlock performance level to set (RFE-REG-001 G9)	RFE-CAL-001 section 5; RFE-STM-001 S14
Throughput	100 kg a shift with a 5 % shredder margin and 2.06 sheets of press capacity for 2	Medium	Shredder and press rates are assumptions inside published ranges; to measure at TRL 4	RFE-CAL-001 section 3; RFE-STM-001 S23, S24
Safety and health	Guarding meets ISO 13857 on paper; hoods sized for 0.5 m/s and the booth extract carries the platen plume about 3.6 times over; noise 79 dB(A) on an assumed sound power	Medium	Air quality and noise can only be shown on a running site; fire load of plastic stock	RFE-STM-001 S10, S15 to S17, S21; RFE-RSK-001
Economics	Breaks even at best: +0.17 USD per shift before rent at a product price of 2.50 USD/kg	Low	Indian listings for regenerated HDPE sheet are about 0.5 to 1 USD/kg; the site needs design-grade products or furniture buyers (RFE-MKT-001)	RFE-CAL-001 section 10
Regulatory	Registration as a plastic waste processor (under 200 t a year band) and Air and Water Act consents in Maharashtra	Medium	Industry category and consent route to assess; local fire and building code review	RFE-REG-001
Supply of parts	Every BOM line priced; machines from open-hardware workspaces or local fabricators	Medium	Single-phase variants of the published 400 V machines are uncommon	`bom/bom.csv`; RFE-MKT-001

Ratings: High, Medium, Low.